

Quiz 4: Qualitative Analysis III — Mixtures & Equations

Review of Session 4. ~15 minutes. A calculator helps for Section C.

Section A: Mixtures

1. A sample fizzes a little with acid and leaves white residue that won't dissolve and is iodine-negative. One powder or a mixture? Of what?
2. You want to separate a salt + cornstarch mixture. Describe the steps.
3. In that separation, which component is in the **filtrate** and which is the **residue**?

Section B: Balancing

Balance by adjusting coefficients only:

1. $\underline{\hspace{1cm}} \text{H}_2 + \underline{\hspace{1cm}} \text{Cl}_2 \rightarrow \underline{\hspace{1cm}} \text{HCl}$
2. $\underline{\hspace{1cm}} \text{Al} + \underline{\hspace{1cm}} \text{O}_2 \rightarrow \underline{\hspace{1cm}} \text{Al}_2\text{O}_3$
3. $\underline{\hspace{1cm}} \text{CH}_4 + \underline{\hspace{1cm}} \text{O}_2 \rightarrow \underline{\hspace{1cm}} \text{CO}_2 + \underline{\hspace{1cm}} \text{H}_2\text{O}$
4. $\underline{\hspace{1cm}} \text{NaHCO}_3 + \underline{\hspace{1cm}} \text{HCl} \rightarrow \underline{\hspace{1cm}} \text{NaCl} + \underline{\hspace{1cm}} \text{H}_2\text{O} + \underline{\hspace{1cm}} \text{CO}_2$
5. Why can't you balance an equation by changing a subscript?

Section C: Moles

1. What is the molar mass of CO_2 ? (C = 12, O = 16)
2. What is the molar mass of NaHCO_3 ? (Na = 23, H = 1, C = 12, O = 16)
3. What is the formula: moles = $\underline{\hspace{2cm}} \div \underline{\hspace{2cm}}$?
4. In a reaction, how do you find the limiting reactant?
5. You have 0.30 mol of reactant A (coefficient 2) and 0.20 mol of reactant B (coefficient 1). Which is limiting?

Answer Key

Section A

1. A mixture — a carbonate (like baking soda) plus an insoluble, non-starch powder (sand or gypsum).
2. Add water and stir (salt dissolves, cornstarch doesn't); filter; evaporate the filtrate to recover the salt; dry the residue on the filter to recover the cornstarch; test each separately.
3. Salt is in the filtrate (dissolved); cornstarch is the residue (on the filter).

Section B

1. 1, 1, 2 2. 4, 3, 2 3. 1, 2, 1, 2 4. 1, 1, 1, 1, 1
2. Changing a subscript changes what the substance *is* (H_2O vs H_2O_2). Only coefficients (the numbers in front) may change.

Section C

1. $12 + 16 + 16 = 44 \text{ g/mol}$.
2. $23 + 1 + 12 + 48 = 84 \text{ g/mol}$.
3. $\text{moles} = \text{mass} \div \text{molar mass}$.
4. Divide each reactant's moles by its coefficient; the smallest result is the limiting reactant.
5. A: $0.30 \div 2 = 0.15$. B: $0.20 \div 1 = 0.20$. 0.15 is smaller → **A is limiting**.